Learning Task-Relevant Gaze for Robust Visual Navigation

Abstract— **Visual navigation policies typically learn to reproduce expert motion from images via behavior cloning (BC). However, standard BC supervises motion prediction without specifying task-relevant image regions, allowing policies to rely on visual correlations that hold within the training distribution but can break down under out-of-distribution (OOD) visual changes. To address this limitation, we propose FoveaNav, a gaze-conditioned navigation framework that is robust under appearance changes. The framework has two modules: A fovea policy selects a gaze location from the current image and a navigation transformer predicts a trajectory from full-image features with cross-attention weighted according to the gaze. Our gaze-conditioning mechanism emphasizes the selected region while preserving peripheral context. Training alternates reinforcement learning updates of the fovea policy with BC updates of the navigation transformer. The fovea policy is rewarded for lower trajectory prediction error relative to expert trajectories, learning where to look without human gaze annotations. We evaluate FoveaNav on five indoor and outdoor real-robot tasks, including target following and obstacle avoidance, under novel distractors, unseen obstacles and background changes. FoveaNav achieves higher closed-loop success rates than full-image BC under OOD visual changes, demonstrating improved navigation robustness.**

I. Introduction

Visual navigation requires identifying task-relevant visual cues: the parts of a scene that provide evidence for the next action [1] (Fig. 1A). The visual evidence needed for action depends on both the current scene conditions and the task. For example, a traffic light or a pedestrian would inform the decision to stop the vehicle. Depending on the task, the relevant cue may be a target to follow, an arrow sign at a junction, or an obstacle together with the free space beside it. Humans focus the visual information needed for action by directing gaze according to the demands of the ongoing task [2]. Inspired by this behavior, we study how a robot can learn task-relevant gaze that adapts to the current scene, selecting the image regions that should inform its next action.

Expert demonstrations are a natural source for learning both where to look and how to move. They can be collected through teleoperation without reward design. Moreover, policies trained on large demonstration datasets have been demonstrated to navigate diverse environments and transfer across robot embodiments [3]–[5]. However, these policies are typically trained with behavior cloning (BC) [6], which supervises the motion the expert produced but not which image regions that motion should depend on. As a result, a policy can fit the demonstrations using visual cues that correlate with actions in the training data without capturing the causal factors that determine those actions. Such reliance on shortcut features can undermine action prediction under visual changes [7], [8]. Rather than changing the training data, we use the expert trajectories to learn not only how to navigate but also where to look. The learned gaze explicitly guides which image regions the policy should emphasize during trajectory prediction.

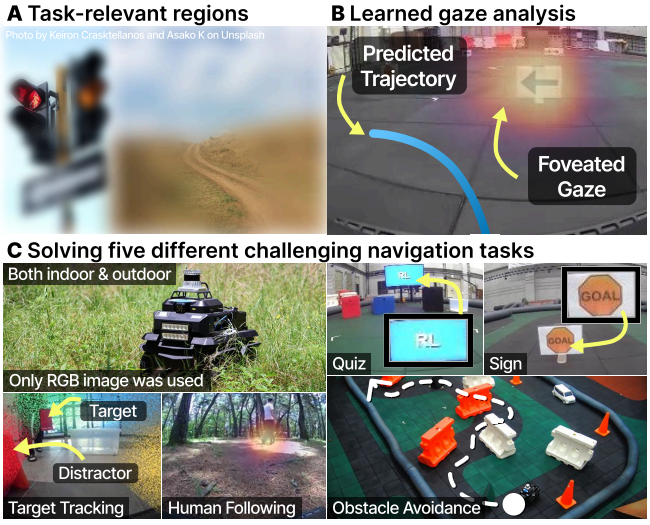


Fig. 1. (A) Different navigation scenarios require different task-relevant visual regions. (B) FoveaNav learns gaze toward these regions and conditions trajectory prediction on the predicted gaze. (C) We evaluate FoveaNav on five indoor and outdoor real-robot navigation tasks.

To this end, we propose **FoveaNav**, a navigation framework that learns a fovea policy for gaze selection based on trajectory prediction accuracy (Fig. 1B). A navigation transformer predicts a future local trajectory from the full image and the selected gaze. The fovea policy receives a higher reward for lower prediction error relative to the expert trajectory. This reward provides supervision for gaze learning without requiring human gaze annotations. Training alternates reinforcement learning (RL) updates of the fovea policy with supervised BC updates of the navigation transformer.

Gaze conditioning should emphasize task-relevant visual regions without losing information needed for navigation. For example, an arrow sign may indicate the desired direction, while surrounding obstacles and free space determine how the robot can move in that direction. A simple approach to gaze-conditioned navigation is to restrict the visual input to a region centered on the selected gaze [9]. However, this discards potentially relevant information outside the selected region. To retain peripheral information, we introduce Gaussian gaze, which emphasizes patches near the selected gaze by reweighting cross-attention over full-image features.

We evaluate FoveaNav on five indoor and outdoor real-robot navigation tasks (Fig. 1C). Our contributions are threefold:

- **Navigation robustness under visual changes** (Section IV-B). We demonstrate that FoveaNav improves navigation robustness under out-of-distribution (OOD)

visual changes compared with BC.

- **Ablation study: gaze learning and conditioning** (Section IV-C). Closed-loop real-robot ablations demonstrate that RL-based gaze learning and Gaussian gaze conditioning contribute to improved navigation performance.
- **Visual sensitivity and gaze behavior** (Section IV-D). Quantitative and qualitative analyses of visual sensitivity and gaze behavior demonstrate that FoveaNav relies on task-relevant image regions for trajectory prediction.

II. Related Work

a) Gaze learning for robot action: Gaze-guided policies have been explored in manipulation with stationary robot bases, where gaze is learned either from human gaze annotations [10], [11] or directly from task supervision. EyeRobot [9], for example, learns gaze through a BC–RL loop that rewards gaze actions yielding accurate action predictions. However, in navigation, robot motion continuously changes the camera viewpoint. As a result, even stationary objects can change position within the image. In addition, we collect demonstrations through third-person teleoperation to give the operator a broader view of the robot's surroundings than the onboard camera provides. This setup does not directly provide human gaze annotations aligned with the robot's images. FoveaNav therefore adopts the BC–RL learning principle to learn which image regions support trajectory prediction, using prediction accuracy as feedback.

b) Gaze conditioning for action prediction: Prior work uses gaze to change the visual input for action prediction. Some methods use gaze-centered crops or multi-resolution images [9], [10], [12], while AGIL [13] processes the original and gaze-weighted images through separate visual streams. More generally, FiLM [14] and gated attention [15] condition visual features on additional inputs. For navigation, gaze conditioning should emphasize task-relevant cues while retaining information about surrounding obstacles and free space needed for trajectory prediction. To meet these requirements, FoveaNav applies Gaussian gaze directly to cross-attention over full-image features. The resulting weights emphasize regions near the selected gaze while preserving access to the rest of the image.

c) Visual analysis: Attention weights alone may not reliably explain model predictions [16]. Prior work therefore uses perturbation-based methods, such as occlusion sensitivity, to identify visual regions that influence predictions [17]. We measure occlusion sensitivity and gaze alignment with task-relevant objects, and apply targeted image corruption to test their influence on trajectory prediction. We also analyze gaze over time, revealing saccade-like shifts between task-relevant regions during navigation.

III. Method

A. Problem Formulation and Overview

We consider visual navigation in which a robot predicts a local trajectory from a front-view RGB image. At time t, the robot observes a front-view RGB image I_t at 320×240 resolution. The navigation transformer predicts a local trajectory $\hat{Y}_t = (\hat{\mathbf{w}}_{t,1}, \ldots, \hat{\mathbf{w}}_{t,J})$ as a sequence of future waypoints. The sequence contains $J = 20$ waypoints at 0.1 s intervals, spanning a two-second prediction horizon. Let $Y_t = (\mathbf{w}_{t,1}, \ldots, \mathbf{w}_{t,J})$ denote the corresponding expert trajectory. The j-th expert waypoint $\mathbf{w}_{t,j}$ is represented as

$$\mathbf{w}_{t,j} = (\Delta x_{t,j}, \Delta y_{t,j}, \cos \Delta\theta_{t,j}, \sin \Delta\theta_{t,j}) \,. \tag{1}$$

Here, $\Delta x_{t,j}$ and $\Delta y_{t,j}$ specify position in the robot's local coordinate frame at time t. The heading offset $\Delta\theta_{t,j}$ is measured relative to the robot's heading at time t.

The fovea policy and navigation transformer share a frozen DINOv3 ViT-B/16 encoder [18]. The encoder maps I_t to N full-image patch tokens $X_t = \{\mathbf{x}_t^i\}_{i=1}^N$, each with 768 dimensions. Using these visual tokens, the fovea policy first selects gaze g_t. The navigation transformer then predicts $\hat{Y}_t = f_\psi(X_t, g_t)$ from the full-image tokens and the selected gaze, where ψ denotes its trainable parameters. We train a separate FoveaNav model for each navigation task. Fig. 2 summarizes the architecture.

B. Gaze-Guided Trajectory Prediction

Let p_i denote the two-dimensional position of patch i in the full-image patch grid. We express gaze g_t in the same coordinate system. The unnormalized Gaussian weight for patch i is $H_{t,i} = \exp(-\|p_i - g_t\|_2^2/(2\sigma^2))$. The parameter σ controls the spatial spread of the Gaussian weights. We set $\sigma = 2$ in patch-grid units. An exponential moving average (EMA) smooths these weights to reduce training instability caused by abrupt changes in gaze. The smoothed weights are normalized to sum to one over all image patches, yielding Gaussian gaze $G_{t,i}$.

The navigation transformer assigns one learnable query to each future waypoint $\{\hat{\mathbf{w}}_{t,j}\}_{j=1}^J$. Each query attends to the current full-image patch tokens through cross-attention. At transformer layer ℓ, we compute the scaled cosine similarity between query vector $\mathbf{q}_j^\ell$ and patch key vector $\mathbf{k}_i^\ell$:

$$S_{j,i}^\ell = \tau^\ell \frac{(\mathbf{q}_j^\ell)^\top \mathbf{k}_i^\ell}{\|\mathbf{q}_j^\ell\|_2 \|\mathbf{k}_i^\ell\|_2}, \tag{2}$$

where τ^ℓ is a learnable scale. A softmax over image patches gives the attention weights $A_{j,i}^\ell = \mathrm{softmax}_i(S_{j,i}^\ell)$.

Gaussian gaze reweights the attention after the softmax. The weights are renormalized over image patches:

$$A_{j,i}'^\ell = \frac{A_{j,i}^\ell \big(G_{t,i} + \epsilon\big)}{\sum_{r=1}^N A_{j,r}^\ell \big(G_{t,r} + \epsilon\big)}. \tag{3}$$

The constant $\epsilon = 10^{-8}$ provides a positive floor for the spatial weights. For each query, these weights determine a weighted sum of value vectors derived from the image tokens. All image tokens remain available during trajectory prediction.

A multilayer perceptron (MLP) decodes each final query representation into a predicted waypoint. The navigation transformer minimizes a Smooth L1 trajectory loss with transition parameter $\beta = 0.5$:

$$\mathcal{L}_{\mathrm{BC}} = \mathrm{SmoothL1}_\beta(\hat{Y}_t, Y_t). \tag{4}$$

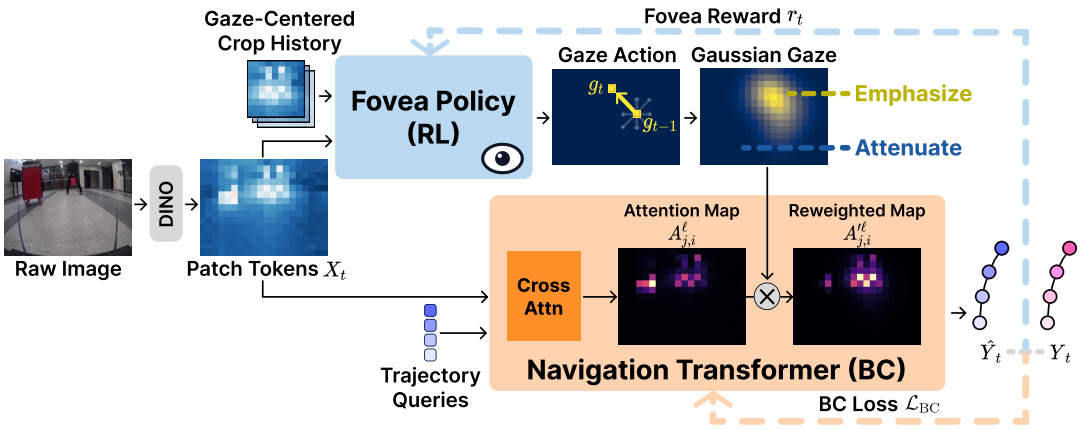


Fig. 2. FoveaNav architecture. The shared visual encoder supplies patch tokens to the fovea policy and navigation transformer. Gaussian gaze reweights cross-attention for trajectory prediction. The dotted path denotes the reward and loss derived from trajectory prediction error.

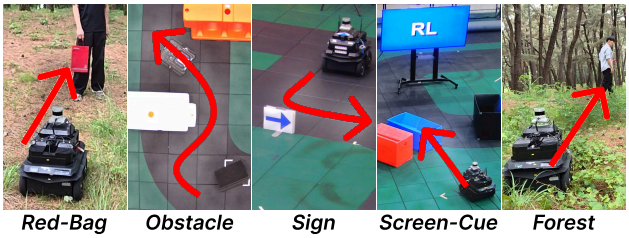


Fig. 3. Representative setups for the five navigation tasks. Red arrows illustrate the intended robot motion.

C. Fovea Policy Training and Robot Control

The fovea policy is a transformer-based actor-critic model. It receives full-image patch tokens, a history of gaze-centered crops from previous frames, the previous gaze action, and grid-position information. The actor selects a discrete action to update gaze g_t on a two-dimensional grid. The action space includes horizontal, vertical, and diagonal movements at multiple step sizes, plus an action that keeps gaze unchanged. Actions that would place the crop outside the image are masked.

We train the fovea policy with proximal policy optimization (PPO) [19]. The reward is a negative weighted sum of trajectory prediction errors:

$$\begin{aligned} r_t = & -\mathrm{ADE}_{xy} - \lambda_{\mathrm{FDE}}\,\mathrm{FDE}_{xy} \\ & - \lambda_{\mathrm{yaw}}\,(\mathrm{AHE} + \lambda_{\mathrm{FHE}}\,\mathrm{FHE})\,. \end{aligned} \tag{5}$$

All errors compare the predicted trajectory $\hat{Y}_t$ with the expert trajectory Y_t. ADE and FDE denote the mean and final Euclidean displacement errors in the local xy plane, respectively. AHE and FHE denote the mean and final heading errors, respectively.

We first pretrain the navigation transformer with full-image BC to stabilize the reward used to train the fovea policy with PPO. At the start of each training iteration, the fovea policy and navigation transformer remain fixed during rollout collection from recorded expert demonstration sequences. We then update only the fovea policy with PPO. After the PPO updates, we update the navigation transformer with BC.

For closed-loop control, an MLP-based inverse dynamics model (IDM) predicts linear and angular velocity commands from the predicted trajectory alone. Implementation and training settings are provided in Section IV-E.

IV. Experiments

We first evaluate navigation robustness through closed-loop trials and open-loop prediction. Next, ablations compare gaze learning and conditioning mechanisms. Finally, visual analyses examine image regions that affect trajectory predictions.

A. Setup and Evaluation Protocols

a) Robot and demonstrations: The experimental platform is a Clearpath Jackal UGV equipped with a front-facing RGB camera. We collect expert demonstrations through joystick teleoperation from a third-person view. Front-camera images are paired with the corresponding robot poses estimated by FAST-LIO2 [20] for trajectory supervision.

b) Tasks and visual changes: We evaluate the five navigation tasks illustrated in Fig. 3. In-distribution (ID) evaluation follows the data-collection distribution. OOD evaluation introduces task-specific visual changes. Demonstration durations are given in parentheses.

Red-Bag Following *(35.73 min).* The robot follows a person carrying a red bag. The demonstrations include obstacle avoidance during target following. We collect demonstrations with the same red bag in varied surroundings. For closed-loop OOD, we introduce other red objects as distractors. For open-loop OOD, we include unseen locations and image corruptions.

Obstacle Avoidance *(39.26 min).* The robot must complete a B-shaped track without colliding with obstacles or leaving the track. The demonstrations include traffic cones, traffic barriers, and chairs. For OOD, we introduce unseen obstacle categories and placements.

Sign-Guided Navigation *(51.84 min).* The robot follows arrow signs at junctions and stops near a goal sign. The demonstrations cover four junction locations and two background configurations. For OOD, we change sign configurations, floor colors, and viewpoints. In some cases, the goal sign is outside the current view.

Screen-Cue Navigation *(27.80 min).* Four boxes colored red, blue, black, and white are arranged in a row. Screen cues instruct the robot to approach the box matching the displayed color. Demonstrations include multiple box orders. We evaluate 28 different screen cues under ID conditions, using a box arrangement seen during training.

TABLE I
CLOSED-LOOP SUCCESS RATES (%) ACROSS FIVE NAVIGATION TASKS.
n: TRIALS PER METHOD.

Task	Setting	n	FoveaNav	BC
Red-Bag	ID	10	100	100
	OOD	30	**90**	20
Obstacle	ID	10	80	80
	OOD	10	**90**	0
Sign	ID	10	100	100
	OOD	55	**96**	38
Screen-Cue	ID	280	**81.8**	31.4
Forest	ID	10	100	100
	OOD	10	**100**	0

Forest Following *(18.97 min).* The robot follows a target person through grassy forest terrain. The target is visible at the start of each sequence. For OOD, we introduce another person closer to the robot than the original target.

c) Baseline: The full-image BC baseline has the same frozen visual encoder, navigation transformer architecture, and trajectory prediction loss as FoveaNav. It predicts trajectories from full-image features without gaze conditioning. The full-image BC baseline receives the same number of gradient updates as the navigation transformer in FoveaNav.

B. Navigation Robustness under Visual Changes

We report success rate (SR) as the fraction of trials satisfying the task objectives described in Section IV-A. FoveaNav selects gaze actions deterministically during evaluation.

FoveaNav achieves higher closed-loop success rates than full-image BC in every evaluated OOD task (Table I). The largest differences occur in *Obstacle Avoidance* and *Forest Following*, where BC fails all OOD trials. FoveaNav also achieves higher success in the ID Screen-Cue setting.

In *Red-Bag Following*, FoveaNav successfully follows the target (Fig. 4). BC instead reaches red distractors. In *Obstacle Avoidance*, FoveaNav completes the track (Fig. 5). BC failures include collisions with unseen obstacles. In *Forest Following*, FoveaNav continues toward the original target (Fig. 6). BC instead follows the other person, who is closer to the robot.

Under ID conditions, FoveaNav and full-image BC achieve identical SR. Beyond SR, we compare completion time, path deviation, and SAM offset to assess execution behavior. FoveaNav achieves lower median completion times across the three tasks shown in Fig. 7. In *Red-Bag Following*, it also achieves lower median path deviation and SAM offset. Path deviation measures how closely the robot follows reference paths collected through expert teleoperation.

Finally, we also compare open-loop prediction performance across six evaluation settings in *Red-Bag Following*. The validation set is randomly held out from the same demonstration dataset for ID evaluation. The metrics are Smooth L1 trajectory loss, ADE_{xy}, and FDE_{xy}. BC achieves lower prediction errors on the clean training and validation sets. By contrast, FoveaNav achieves lower errors under unseen-location and corrupted-image conditions (Fig. 8).

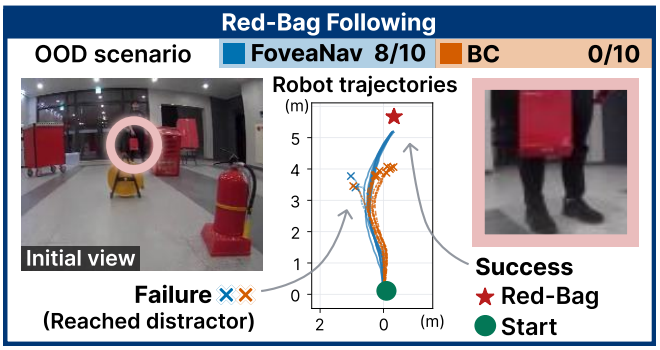


Fig. 4. Closed-loop *Red-Bag Following* with red distractors. Counts report successful trials out of total trials in the illustrated setting.

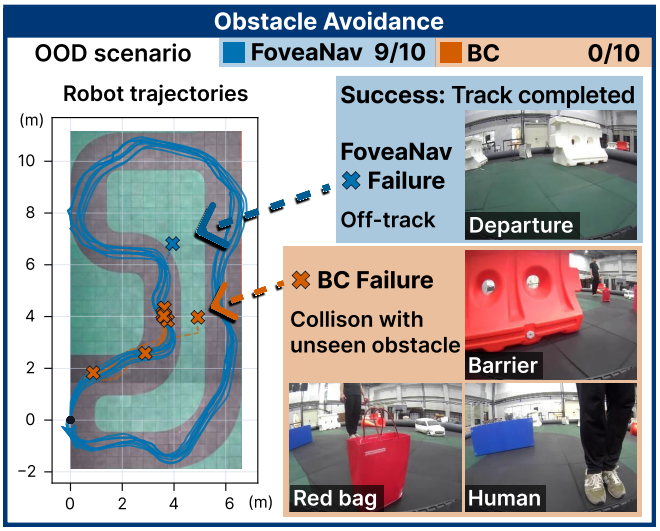


Fig. 5. Closed-loop *Obstacle Avoidance* with unseen obstacle categories and placements.

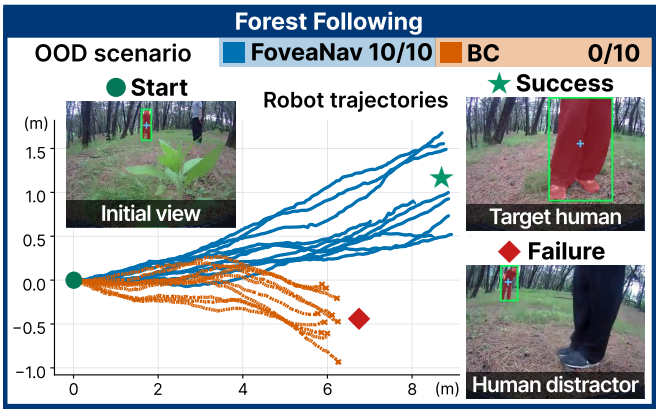


Fig. 6. Closed-loop *Forest Following* with an additional person acting as a human distractor, who is closer to the robot than the target human.

C. Ablation Study: Gaze Learning and Conditioning

We evaluate how gaze learning and conditioning affect visual navigation. FoveaNav combines RL-based gaze learning with Gaussian gaze conditioning. We compare these design choices in closed-loop real-robot trials on *Red-Bag Following* with and without obstacles.

To evaluate RL-based gaze learning, we construct an externally supervised baseline that learns to direct gaze toward the target. Our demonstrations do not include human gaze annotations, so we train a target-supervised fovea policy using red-bag masks generated by SAM 3 [21]. On held-out data, its

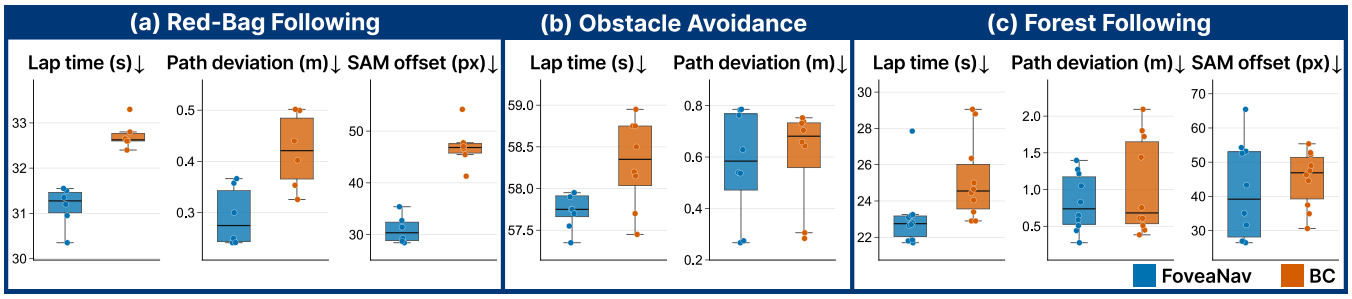


Fig. 7. Closed-loop execution metrics under ID conditions for *Red-Bag Following*, *Obstacle Avoidance*, and *Forest Following*. Lap time denotes task completion time. SAM offset is the pixel distance between the image center and the centroid of the target mask obtained with SAM 3 [21]. The target is the red bag in *Red-Bag Following* and the person being followed in *Forest Following*. This metric measures how well the robot keeps the target centered in the image while following it. All metrics include successful trials only. Lower values are better.

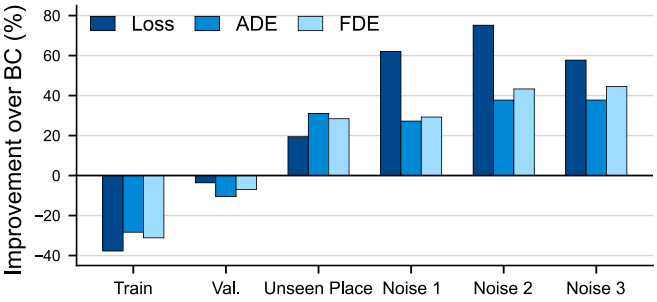


Fig. 8. Relative reduction in open-loop prediction errors compared with full-image BC. Each percentage expresses the error reduction relative to the corresponding BC error. Positive values indicate lower errors for FoveaNav. Noise 1–3 indicate increasing corruption severity.

112×112 gaze crop contains an average of 99.1% of the target-mask pixels per frame. This confirms high target coverage for the target-supervised baseline. To evaluate Gaussian gaze conditioning, we compare it with hard-crop conditioning, which restricts the navigation transformer's visual input to a gaze-centered crop.

We compare FoveaNav (Ours) with four variants (Table II). Variant A follows the same BC–RL training procedure as FoveaNav but replaces Gaussian gaze with hard-crop inputs. Variant B applies Gaussian gaze using the target-supervised fovea policy described above. We additionally examine whether further RL training improves navigation when gaze has already been learned through target supervision. Starting from the same target-supervised fovea policy, we train variants C and D through the BC–RL loop with Gaussian gaze and hard-crop inputs, respectively.

The results show that all three variants with Gaussian gaze conditioning succeed without obstacles (B, C, and Ours). With obstacles, the variant further trained through the BC–RL loop achieves higher success than the variant using the target-supervised fovea policy directly (B vs. C). Furthermore, RL training without target supervision achieves the highest success (C vs. Ours). These results indicate that accurate target localization alone is insufficient for reliable following with obstacle avoidance, and that RL-based gaze learning improves navigation in this setting.

The conditioning comparisons also favor Gaussian gaze in trials with obstacles. After target-supervised training followed by RL, Gaussian gaze achieves higher success than hard-crop inputs (C vs. D). The same advantage holds when gaze is learned without target supervision (A vs. Ours). The hard-crop variant trained without target supervision also fails without obstacles. Inspection of its gaze behavior revealed cases in which the target was initially outside the crop and the policy failed to bring it into the crop during the trial. With target-supervised pretraining, the hard-crop variant achieves higher success in the same setting. Together, these comparisons demonstrate the benefits of RL-based gaze learning and Gaussian gaze conditioning for target following with obstacle avoidance.

TABLE II

CLOSED-LOOP *Red-Bag Following* WITH AND WITHOUT OBSTACLES.

Variant	Gaze training	Conditioning	With obstacles	Without obstacles
Ours	RL	Gaussian gaze	**10/10**	**10/10**
A	RL	Hard crop	1/10	0/10
B	Supervised	Gaussian gaze	5/10	**10/10**
C	Supervised → RL	Gaussian gaze	8/10	**10/10**
D	Supervised → RL	Hard crop	1/10	7/10

D. Visual Sensitivity and Gaze Behavior

We next examine which image regions influence the predictions of these models and how the learned gaze changes during navigation.

a) Visual sensitivity: We first quantify the alignment between the most occlusion-sensitive image region and the task-relevant object (the red bag in this evaluation; Fig. 9). We then visualize occlusion sensitivity alongside gaze, attention, and predicted trajectories (Fig. 10). Finally, we compare trajectory prediction errors when image corruption preserves or also affects the task-relevant object (Fig. 11).

For the quantitative analysis, we scan each image using patches of 32×32 pixels at a stride of 32 pixels. Each perturbation replaces one patch with the image-mean RGB color. We pass each perturbed image through the complete model to predict a trajectory. Occlusion sensitivity is measured as the average distance between corresponding xy waypoints in the original and perturbed predictions. For each image, we identify the patch whose occlusion causes the largest change in the predicted trajectory. We then measure the distance in pixels between the center of this patch and the centroid of the target mask obtained with SAM. We refer to this metric as *occlusion-to-target distance*.

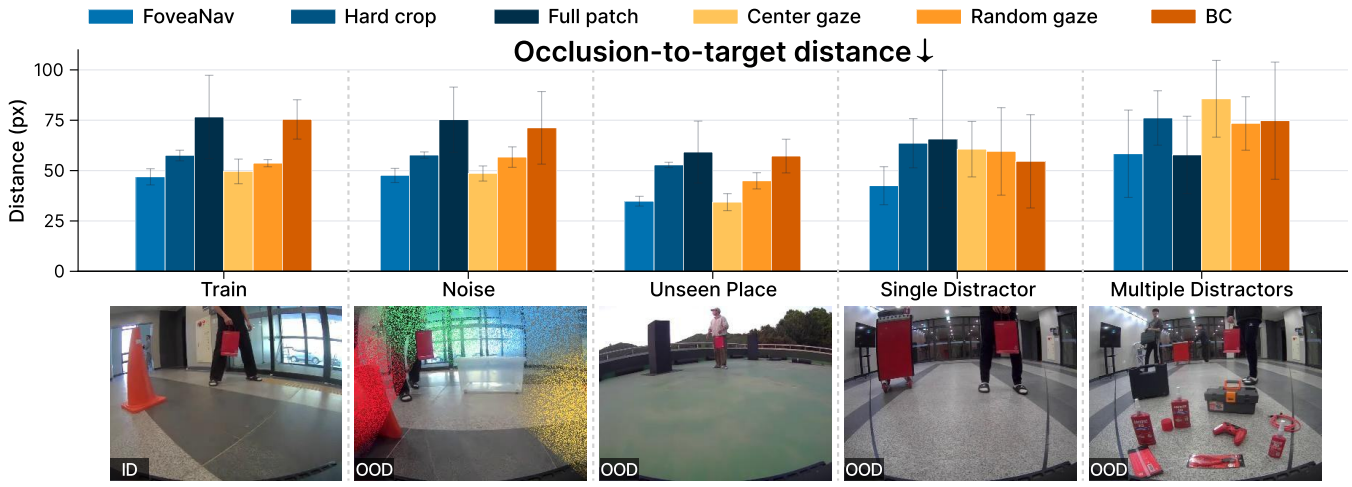


Fig. 9. Occlusion-to-target distance on *Red-Bag Following*. The distance is measured between the center of the most occlusion-sensitive patch and the centroid of the red-bag. Lower values indicate that the most sensitive patch lies closer to the red bag. Example images illustrate the evaluation conditions.

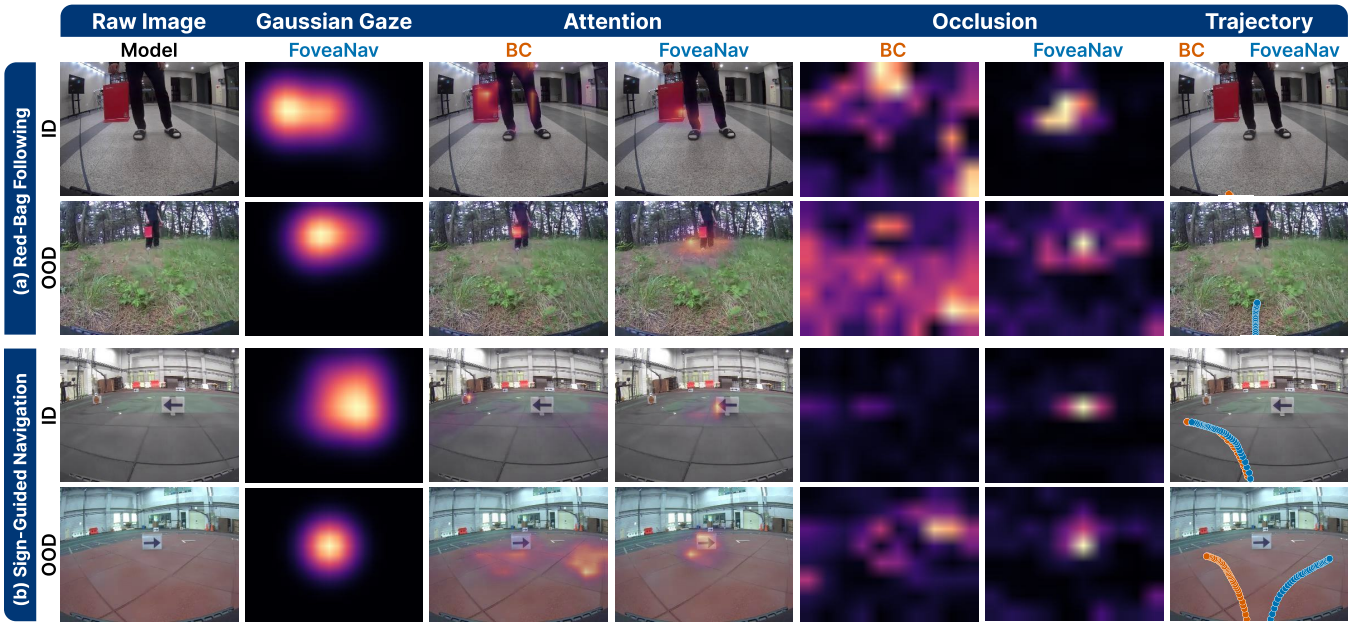


Fig. 10. Qualitative comparisons under ID and OOD conditions in *Red-Bag Following* and *Sign-Guided Navigation*. Columns present the input image, Gaussian gaze, attention overlays, occlusion sensitivity maps, and predicted trajectories. Attention maps are averaged over navigation transformer layers, attention heads, and waypoint queries. Occlusion maps indicate how strongly masking each image patch changes the predicted trajectory. Orange and blue curves denote full-image BC and FoveaNav trajectories, respectively.

We compare the following variants using occlusion-to-target distance. The *Center gaze* and *Random gaze* variants set gaze to the image center or a randomly selected valid position, respectively. Both retain Gaussian gaze conditioning. The *Hard crop* and *Full patch* variants evaluate alternative conditioning mechanisms for learned gaze. The *Hard crop* variant predicts trajectories from gaze-centered crops. The *Full patch* variant jointly processes gaze-patch tokens and full-image tokens without Gaussian gaze. Full-image BC provides a reference without gaze conditioning. We evaluate these variants across five ID and OOD settings in *Red-Bag Following*. FoveaNav achieves the lowest occlusion-to-target distance in most conditions (Fig. 9). These results indicate closer alignment between FoveaNav's most occlusion-sensitive image regions and the task-relevant object.

To interpret these quantitative results, we visualize occlusion sensitivity in *Red-Bag Following* and *Sign-Guided Navigation* (Fig. 10). Gaussian gaze is concentrated near the red bag or arrow sign in these examples. BC also assigns high attention weights to these task cues. However, its occlusion sensitivity extends more broadly into surrounding image regions than that of FoveaNav. The corresponding trajectory predictions also differ. In the OOD sign example, FoveaNav predicts a right turn consistent with the arrow sign. BC instead predicts a left turn.

We extend this qualitative analysis to all five navigation tasks. Additional visualizations are provided in the accompanying video.

We further examine how trajectory loss changes when corruption also affects the target (Fig. 11). Target-preserving corruption adds noise only outside the target mask. Full-image corruption also affects pixels inside the target mask. FoveaNav

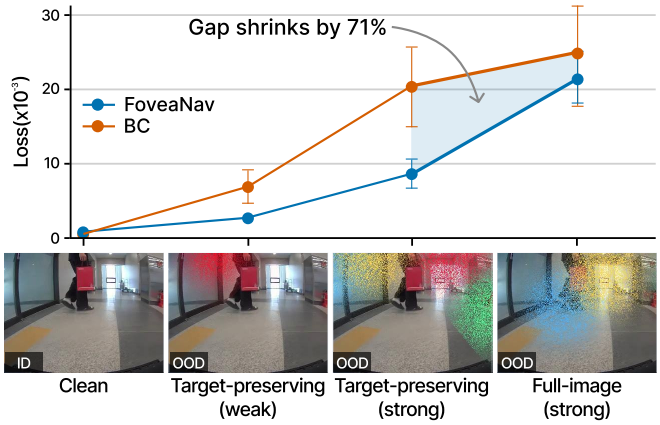


Fig. 11. Smooth L1 trajectory loss under image corruption in *Red-Bag Following*. The percentage indicates the reduction in the BC–FoveaNav loss gap under strong full-image corruption relative to strong target-preserving corruption.

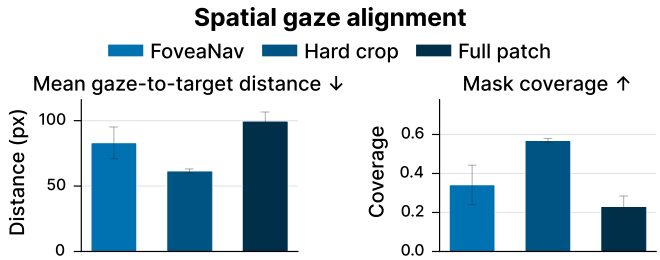


Fig. 12. Spatial gaze alignment across conditioning methods on *Red-Bag Following*. Bars and error bars indicate the mean and standard deviation across three seeds.

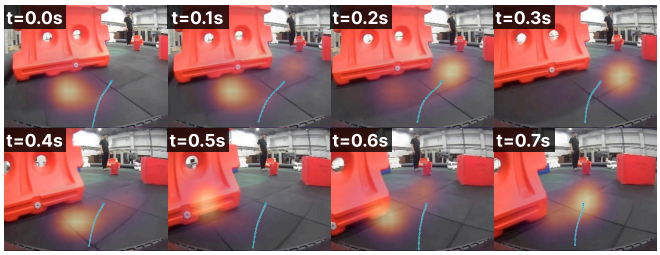


Fig. 13. EMA-smoothed Gaussian gaze in *Obstacle Avoidance*. Temporal smoothing retains spatial weights near the previous gaze. The resulting distribution temporarily spans both the obstacle and the traversable gap during gaze transitions.

maintains a lower loss than BC as target-preserving corruption increases from weak to strong. Under strong full-image corruption, FoveaNav's loss is closer to that of BC, with a 71% smaller gap than under strong target-preserving corruption. Together with the occlusion results, this finding supports FoveaNav's reliance on task-relevant visual information for trajectory prediction.

b) Gaze alignment and temporal behavior: We examine where the learned gaze is directed and how it changes during navigation. We first evaluate how well the predicted gaze aligns with the red bag under each conditioning method. The comparison includes FoveaNav and the *Hard crop* and *Full patch* variants introduced in the occlusion analysis (Fig. 12).

Mean gaze-to-target distance measures the average pixel distance between the predicted gaze and the target-mask centroid. Mask coverage measures the fraction of target-mask pixels within a 112×112 gaze-centered crop. *Hard crop* achieves the lowest mean gaze-to-target distance and the highest mask coverage. These results indicate that the learned gaze is more concentrated on the target under hard-crop conditioning. However, FoveaNav achieves higher success than *Hard crop* in the preceding closed-loop trials with obstacles. It also achieves lower occlusion-to-target distances in most conditions of the occlusion analysis. Together, these results indicate that target alignment alone does not explain the observed navigation performance. Navigation with obstacle avoidance also requires visual information about surrounding obstacles and traversable space.

We next examine the temporal behavior of FoveaNav in *Obstacle Avoidance* (Fig. 13). Gaze alternates between a nearby obstacle and a narrow traversable gap in saccade-like shifts.

TABLE III

GAUSSIAN GAZE SCALE ABLATION ON 2,000 OOD FRAMES.

Setting	σ	Loss	ADE	FDE
Fixed	1	0.0362	0.1030	0.2174
Fixed	2	**0.0129**	**0.0680**	**0.1356**
Fixed	5	0.0153	0.1030	0.2056
Learnable	$2 \rightarrow 4.37$	0.0159	0.0848	0.1835

TABLE IV

BC ARCHITECTURE ABLATION ON THE VALIDATION SPLIT.

Attention	Score	Loss	ADE	FDE
Cross-attention	Scaled cosine	**0.0069**	**0.0282**	**0.0496**
Cross-attention	Dot product	0.0090	0.0363	0.0645
Self-attention	Scaled cosine	0.0087	0.0356	0.0620
Self-attention	Dot product	0.0092	0.0400	0.0697

TABLE V

OPEN-LOOP VALIDATION LOSS ON EYEGYM.

Task	AVP	EyeRobot	FoveaNav360
Towel	Off	0.4330	**0.3484**
	On	0.4406	**0.3472**
Brush	Off	0.4140	**0.3407**
	On	0.4283	**0.3401**

E. Additional Ablations and Experimental Details

a) Gaussian gaze scale: The fixed scale $\sigma = 2$ achieves the lowest loss, ADE, and FDE among the evaluated settings (Table III).

b) BC architecture ablation: Cross-attention with scaled cosine scores achieves the lowest validation loss, ADE, and FDE (Table IV). With 20 trajectory queries and 300 visual tokens, cross-attention reduces attention matrix-multiplication cost by 94.1% relative to joint self-attention.

c) EyeGym comparison: We adapt FoveaNav to panoramic images (FoveaNav360) and evaluate it in EyeGym, which simulates camera views from recorded manipulation demonstrations [9]. With Gaussian gaze, FoveaNav360 achieves lower validation loss than EyeRobot across all evaluated tasks, both with and without Active Visual Pretraining (AVP) (Table V).

d) Data split: We split each task's demonstrations into training and validation sets at a 9:1 ratio. The split unit is a five-second chunk for *Red-Bag Following*, *Obstacle Avoidance*, and *Forest Following*, or a complete sequence for the remaining tasks.

e) Model and training settings: The navigation transformer uses four layers, four attention heads per layer, a feed-forward dimension of 3072, and dropout of 0.1. The attention scale τ^{ℓ} starts at 3 and is capped at 6. The EMA assigns coefficients of 0.7 and 0.3 to the previous smoothed and current Gaussian weights, respectively. The fovea policy uses three previous 112×112 gaze-centered crops and a 14×9 gaze grid with 16-pixel spacing. Gaze step sizes are 1, 2, and 4 grid cells.

BC pretraining lasts 100 epochs. PPO uses 20-step rollouts and ten update epochs with two minibatches. The discount factor and generalized advantage estimation parameter are 0.99 and 0.95, respectively. The clipping, value-loss, and entropy coefficients are 0.2, 0.5, and 0.05, respectively. Reward weights are $\lambda_{\mathrm{FDE}} = 0.5$, $\lambda_{\mathrm{yaw}} = 1$, and $\lambda_{\mathrm{FHE}} = 0.5$.

Learning rates are 10^{-3} for the navigation transformer and 5×10^{-4} for the fovea policy. Training uses weight decay of 10^{-4}, gradient clipping at 1.0, and mixed precision on 8 NVIDIA A100 GPUs with 80 rollout environments per GPU.

f) Robot control and inference time: The IDM is trained separately on task-specific trajectory–command pairs with a Smooth L1 loss. For each task, FoveaNav and full-image BC share the same IDM in closed-loop evaluation. FoveaNav runs at 10 Hz on an onboard NVIDIA RTX 3060 GPU.

On an NVIDIA RTX 3080 Ti, inference latency is 35.9 ms for FoveaNav, compared with 23.4 ms for full-image BC. The implementation retains full-image feature computation.

V. Discussion and Conclusion

We presented FoveaNav, a framework that learns task-relevant gaze for trajectory prediction. In real-robot experiments, FoveaNav achieved substantially higher success rates than full-image BC under OOD visual changes. BC nevertheless achieved lower prediction errors in ID open-loop evaluation. These findings highlight the importance of explicitly selecting task-relevant visual information for navigation robustness beyond prediction accuracy on the training distribution. Real-robot ablations further demonstrated the benefits of RL-based gaze learning and Gaussian gaze conditioning. Quantitative analyses and visualizations supported FoveaNav's reliance on task-relevant image regions for trajectory prediction. Spatial and temporal gaze analyses further indicated that task-relevant visual information extends beyond a single target object and varies with the scene.

The current setup relies on observations from a camera fixed to the robot. This limits the robot's ability to respond to environmental changes outside the camera's field of view. The method also focuses on short-horizon trajectory prediction. Future work includes integrating local trajectory prediction with long-horizon planning for more complex navigation tasks. Another research direction is to extend learned visual selection and conditioning to other image-based prediction problems.

References

[1] M. Nazeri, J. Wang, A. Payandeh, and X. Xiao, "VANP: Learning where to see for navigation with self-supervised vision-action pre-training," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024, pp. 2741–2746.

[2] M. Hayhoe and D. Ballard, "Eye movements in natural behavior," *Trends in Cognitive Sciences*, vol. 9, no. 4, pp. 188–194, 2005.

[3] D. Shah, A. Sridhar, A. Bhorkar, N. Hirose, and S. Levine, "GNM: A general navigation model to drive any robot," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2023, pp. 7226–7233.

[4] D. Shah, A. Sridhar, N. Dashora, K. Stachowicz, K. Black, N. Hirose, and S. Levine, "ViNT: A foundation model for visual navigation," in *Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 229, 2023, pp. 711–733.

[5] A. Sridhar, D. Shah, C. Glossop, and S. Levine, "NoMaD: Goal masked diffusion policies for navigation and exploration," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2024, pp. 63–70.

[6] Y. LeCun, U. Muller, J. Ben, E. Cosatto, and B. Flepp, "Off-road obstacle avoidance through end-to-end learning," in *Advances in Neural Information Processing Systems*, vol. 18, 2005.

[7] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, "Shortcut learning in deep neural networks," *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020.

[8] L. Gong, F. Bahrani, Y. Zhou, A. Banayeeanzade, J. Li, and E. Bıyık, "AutoFocus-IL: VLM-based saliency maps for data-efficient visual imitation learning without extra human annotations," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[9] J. Kerr, K. Hari, E. Weber, C. M. Kim, B. Yi, t. bonnen, K. Goldberg, and A. Kanazawa, "Eye, Robot: Learning to look to act with a BC-RL perception-action loop," in *Conference on Robot Learning (CoRL)*, ser. Proceedings of Machine Learning Research, vol. 305, 2025, pp. 3647–3664.

[10] H. Kim, Y. Ohmura, and Y. Kuniyoshi, "Using human gaze to improve robustness against irrelevant objects in robot manipulation tasks," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 4415–4422, 2020.

[11] R. Takizawa, I. Karino, K. Nakagawa, Y. Ohmura, and Y. Kuniyoshi, "Enhancing reusability of learned skills for robot manipulation via gaze information and motion bottlenecks," *IEEE Robotics and Automation Letters*, vol. 10, no. 10, pp. 10 737–10 744, 2025.

[12] I. Chuang, J. Zou, A. Lee, D. Gao, and I. Soltani, "Look, focus, act: Efficient and robust robot learning via human gaze and foveated vision transformers," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2026.

[13] R. Zhang, Z. Liu, L. Zhang, J. A. Whritner, K. S. Muller, M. M. Hayhoe, and D. H. Ballard, "AGIL: Learning attention from human for visuomotor tasks," in *European Conference on Computer Vision (ECCV)*, 2018, pp. 663–679.

[14] E. Perez, F. Strub, H. de Vries, V. Dumoulin, and A. Courville, "FiLM: Visual reasoning with a general conditioning layer," in *AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 3942–3951.

[15] D. S. Chaplot, K. Mysore Sathyendra, R. K. Pasumarthi, D. Rajagopal, and R. Salakhutdinov, "Gated-attention architectures for task-oriented language grounding," in *AAAI Conference on Artificial Intelligence*, vol. 32, no. 1, 2018, pp. 2819–2826.

[16] S. Jain and B. C. Wallace, "Attention is not explanation," in *2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, 2019, pp. 3543–3556.

[17] M. D. Zeiler and R. Fergus, "Visualizing and understanding convolutional networks," in *European Conference on Computer Vision (ECCV)*, ser. Lecture Notes in Computer Science, vol. 8689, 2014, pp. 818–833.

[18] O. Siméoni, H. V. Vo, M. Seitzer, F. Baldassarre, M. Oquab, C. Jose, V. Khalidov, M. Szafraniec, S. E. Yi, M. Ramamonjisoa, *et al.*, "DINOv3," *Transactions on Machine Learning Research*, 2026.

[19] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, and O. Klimov, "Proximal policy optimization algorithms," *arXiv preprint arXiv:1707.06347*, 2017.

[20] W. Xu, Y. Cai, D. He, J. Lin, and F. Zhang, "FAST-LIO2: Fast direct LiDAR-inertial odometry," *IEEE Transactions on Robotics*, vol. 38, no. 4, pp. 2053–2073, 2022.

[21] N. Carion, L. Gustafson, Y.-T. Hu, S. Debnath, R. Hu, D. Suris Coll-Vinent, C. Ryali, K. V. Alwala, H. Khedr, A. Huang, *et al.*, "SAM 3: Segment anything with concepts," in *International Conference on Learning Representations (ICLR)*, 2026, pp. 138 846–138 923.